

Practical No. 02

ON

SSCA4030 - Image Processing

TITLE: Conversion of Images to Grayscale and Different Color Models

BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)

SUBMITTED BY:

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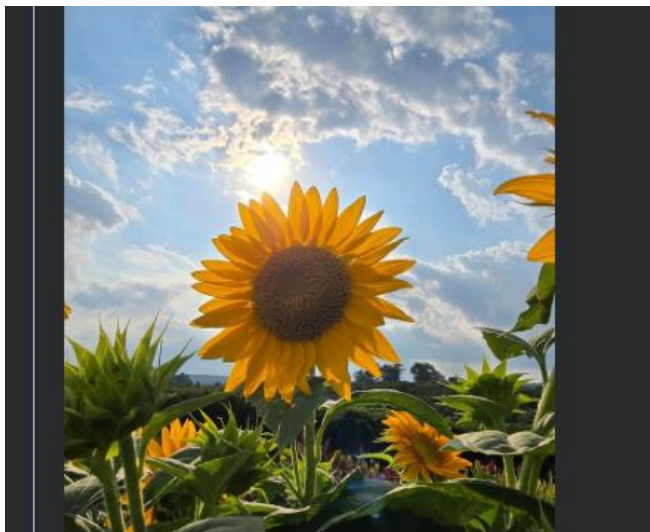
Aim: To understand and implement the conversion of a colour image into grayscale and various colour models such as RGB, HSV, LAB, and YCrCb using Python and OpenCV.

Input Code:

```
# Import required libraries
import cv2
import matplotlib.pyplot as plt
from google.colab.patches import cv2_imshow

# Read image
image = cv2.imread("/content/Flower.jpg")

# Check if image loaded successfully
if image is None:
    print("Error: Flower.jpg not found.")
else:
    cv2_imshow(image)
```



```
# Convert BGR to RGB for display
rgb_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
cv2_imshow(rgb_image)
```



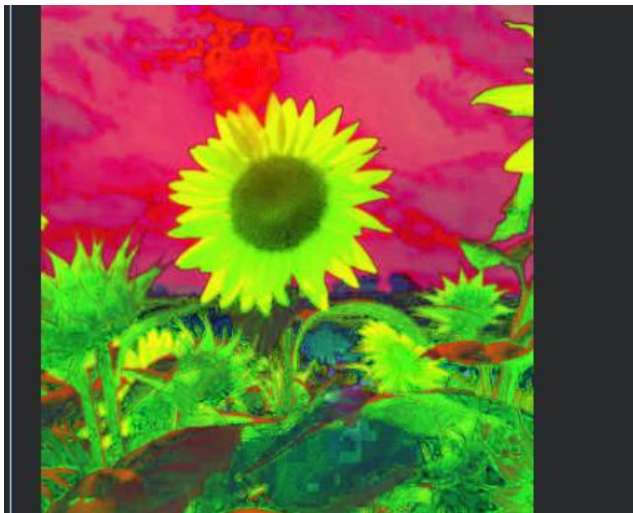
Convert to Grayscale

```
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)  
cv2.imshow(gray_image)
```



Convert to HSV

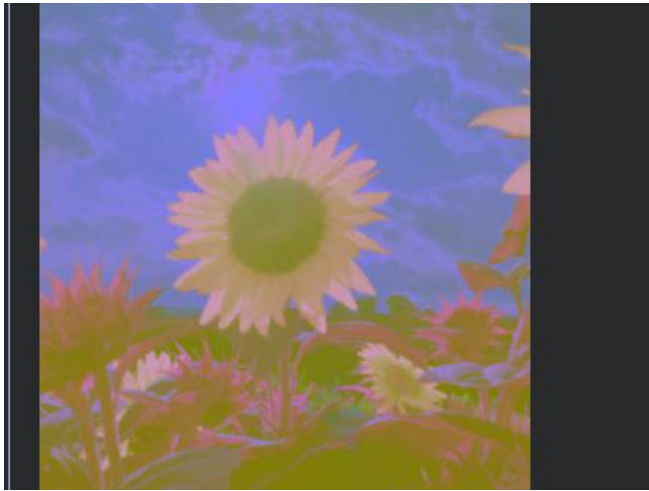
```
hsv_image = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)  
cv2.imshow(hsv_image)
```



```
# Convert to LAB
```

```
lab_image = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)
```

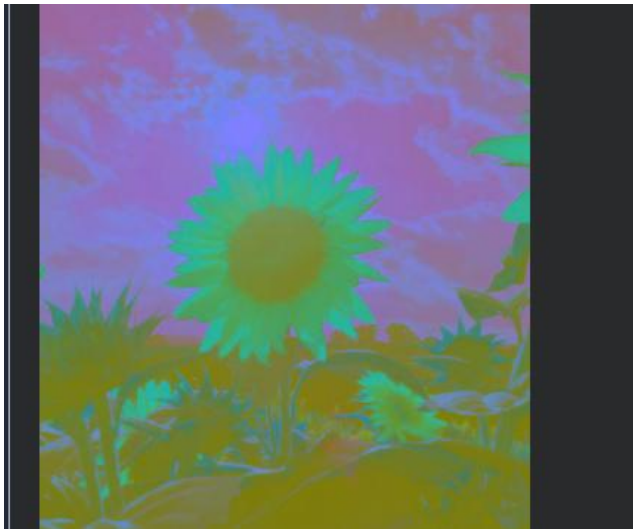
```
cv2.imshow('lab_image')
```



```
# Convert to YCrCb
```

```
ycrcb_image = cv2.cvtColor(image, cv2.COLOR_BGR2YCrCb)
```

```
cv2.imshow('ycrcb_image')
```



```
# Display images
```

```
titles = [
```

```
    "Original RGB",
```

```
    "Grayscale",
```

```
    "HSV",
```

```
    "LAB",
```

```
    "YCrCb"
```

```
]
```

```

images = [
    rgb_image,
    gray_image,
    hsv_image,
    lab_image,
    ycrb_image
]

plt.figure(figsize=(15, 8))
for i in range(len(images)):
    plt.subplot(2, 3, i + 1)
    if i == 1:
        plt.imshow(images[i], cmap="gray")
    else:
        plt.imshow(images[i])
    plt.title(titles[i])
    plt.axis("off")

plt.tight_layout()
plt.show()

```



Questions

Q1. Convert an image into the CMYK color model and explain the process.

Answer: CMYK stands for **Cyan, Magenta, Yellow, and Key (Black)**. It is mainly used in color printing. The RGB image is converted into CMYK by calculating the Cyan, Magenta, Yellow, and Black values from the RGB components. CMYK provides better color reproduction for printed materials.

Q2. Compare RGB and HSV using suitable examples.

Answer:

RGB

Uses Red, Green, Blue components

Suitable for image display

Difficult to separate colors

Example: Computer screens

HSV

Uses Hue, Saturation, Value

Suitable for color detection

Easy to identify colors

Example: Traffic light detection

Q3. Perform color-based object detection using the HSV color space.

Answer: In HSV color space, a specific color range is selected using Hue values. A mask is created to highlight only the desired color, and objects having that color are detected. HSV is widely used for object tracking and color segmentation.

Q4. Extract only the luminance channel from a YCrCb image.

Answer:

In the YCrCb color model, the **Y channel** represents luminance (brightness). To extract luminance, the image is converted to YCrCb format and only the Y channel is selected. This channel contains brightness information without color details.

Q5. Plot histograms for grayscale and RGB images and compare them.

Answer: A histogram shows the distribution of pixel intensities in an image.

- **Grayscale Histogram:** Displays intensity values from black (0) to white (255).
- **RGB Histogram:** Displays separate distributions for Red, Green, and Blue channels.